

Profile-Based Matching Algorithm

Project Submission Report:

Student Name: Keshav Vittal Marathe

Batch: July 2026

Project Title:

Intelligent Hybrid Recommendation System – Profile-Based Matching Algorithm

1. Data Pipeline Script:

Users.csv dataset is created in google excel sheet manually not by python code. After creating 100 users data, functions like cleaning data, mbti matching score, location score are created and done. The features included in the **datasets** are:

user_id
Name
age
gender
city
profession
industry
experience
education
preferred_work_mode
skills
career_goal
mbti
interests
languages
profession_summary
about_me

Data Cleaning Process:

```
def clean_text(text):
    text = text.lower()

    # Remove punctuation and numbers
    text = ''.join(char for char in text if char.isalpha() or char.isspace())

    # Tokenization
    words = text.split()

    # Remove stopwords and lemmatize
    words = [
        lemmatizer.lemmatize(word)
        for word in words
        if word not in stop_words
    ]

    return " ".join(words)
```

Matching Algorithm:

Recommendation Function Demonstration: To evaluate the effectiveness of the adaptive feedback mechanism, the recommendation system is demonstrated using the same **User ID** in two stages: **before adaptive learning** and **after adaptive learning**. The first result is generated using the initial hybrid weighting scheme, while the second uses the updated weights learned from user feedback. Comparing these two outputs illustrates how the adaptive learning process refines compatibility scores and improves the relevance of the recommended matches.

Top 5 User Recommendations for User ID “U011” Before Adaptive Learning:

```
selected_user = "U011"

print("Selected User Details:\n")

display(
    df[df["user_id"] == selected_user][["user_id",
    "name",
    "profession",
    "city",
    "mbti"
    ]]
)

print("\nTop 5 Recommended Users:\n")

display(recommend_users(selected_user))
```

Selected User Details:

	user_id	name	profession	city	mbti
10	U011	Karan Patel	Data Scientist	Kochi	ISTJ

Top 5 Recommended Users:

	User ID	Name	Profession	City	MBTI	Text Similarity (%)	MBTI Score	Location Score	Compatibility Score
0	U001	Aarav Verma	Data Scientist	Mumbai	ISTJ	95.11	90	50	84.56
89	U091	Nikhil Kulkarni	Data Scientist	Hyderabad	INTJ	97.60	75	50	81.30
19	U021	Akash Rao	Data Scientist	Bengaluru	INTJ	95.95	75	50	80.47
29	U031	Ishita Khan	Data Scientist	Jaipur	INTP	95.41	75	50	80.20
79	U081	Sneha Das	Data Scientist	Ahmedabad	ISTJ	72.92	90	50	73.46

Top 5 Recommendations for User ID “U011” After Adaptive Learning:

```
selected_user = "U011"

print("Selected User Details:\n")

display(
    df[df["user_id"] == selected_user][["user_id",
    "name",
    "profession",
    "city",
    "mbti"
    ]]
)

print("\nTop 5 Recommended Users:\n")

display(recommend_users_after_learning(selected_user))
```

Selected User Details:

	user_id	name	profession	city	mbti
10	U011	Karan Patel	Data Scientist	Kochi	ISTJ

Top 5 Recommended Users:

	User ID	Name	Profession	City	MBTI	Text Similarity (%)	MBTI Score	Location Score	Compatibility Score
0	U001	Aarav Verma	Data Scientist	Mumbai	ISTJ	95.11	90	50	78.61
89	U091	Nikhil Kulkarni	Data Scientist	Hyderabad	INTJ	97.60	75	50	73.80
19	U021	Akash Rao	Data Scientist	Bengaluru	INTJ	95.95	75	50	73.29
29	U031	Ishita Khan	Data Scientist	Jaipur	INTP	95.41	75	50	73.13
75	U077	Aisha Das	DevOps Engineer	Kochi	ISTJ	22.18	90	100	72.57

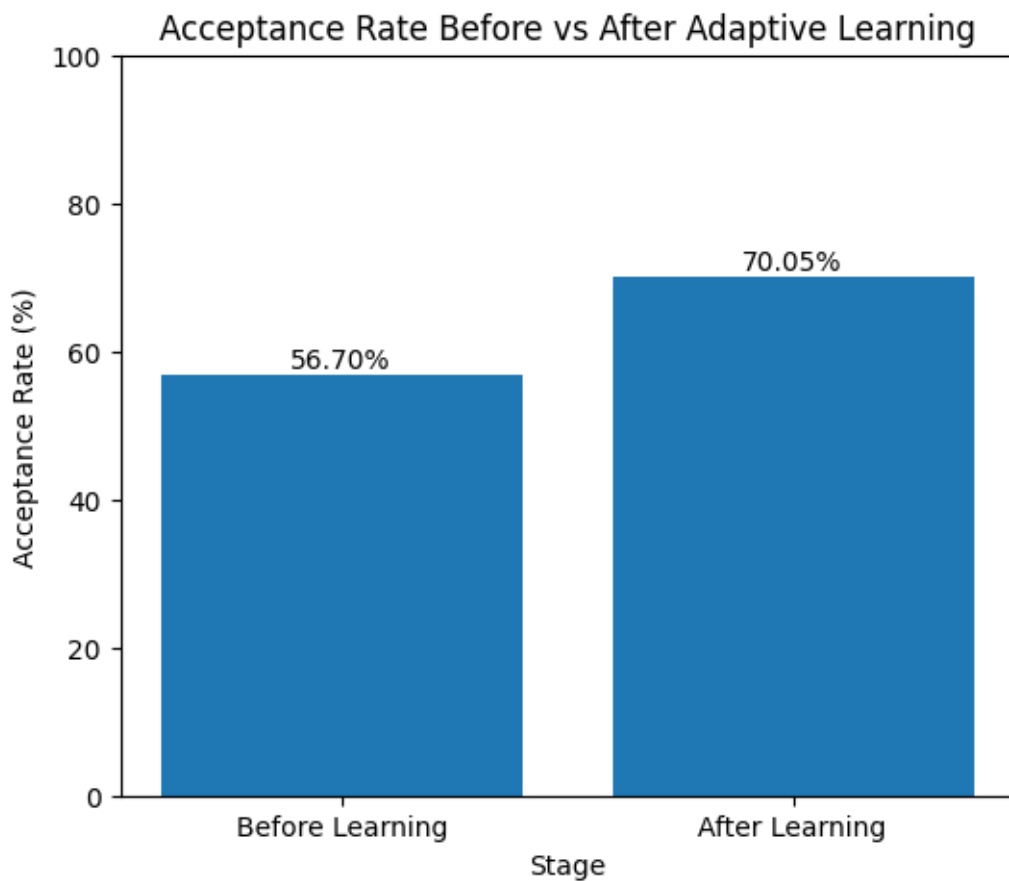
Performance Analysis:

The performance of the hybrid recommendation system was evaluated by comparing the acceptance rate before and after applying the adaptive feedback mechanism. User interactions (Accept/Reject) were used to update the recommendation weights, and the system was evaluated again to measure the improvement in recommendation quality.

Result Table:

Stage	Acceptance rate
Before Adaptive Learning	56.70%
After Adaptive Learning	70.05%

Performance Graph:



UI Demo:

